

Article

Multimodal Food Image Classification with Large Language Models

Jun-Hwa Kim ^{1,*} , Nam-Ho Kim ² , Donghyeok Jo ³  and Chee Sun Won ³ ¹ Department of Artificial Intelligence, Konyang University, Daejeon 35365, Republic of Korea² Korean Broadcasting System, Seoul 07235, Republic of Korea; namho96@kbs.co.kr³ Department of Electronics and Electrical Engineering, Dongguk University, Seoul 04620, Republic of Korea; jdh1184@dgu.ac.kr (D.J.); cswon@dongguk.edu (C.S.W.)

* Correspondence: junhwakim@konyang.ac.kr

Abstract: In this study, we leverage advancements in large language models (LLMs) for fine-grained food image classification. We achieve this by integrating textual features extracted from images using an LLM into a multimodal learning framework. Specifically, semantic textual descriptions generated by the LLM are encoded and combined with image features obtained from a transformer-based architecture to improve food image classification. Our approach employs a cross-attention mechanism to effectively fuse visual and textual modalities, enhancing the model's ability to extract discriminative features beyond what can be achieved with visual features alone.

Keywords: food image classification; fine-grained visual classification; multimodal image feature; large language model; deep learning

1. Introduction

In recent years, large language models (LLMs) have attracted significant attention in artificial intelligence (AI), extending their influence beyond natural language processing to fields such as computer vision [1,2]. One domain where LLMs are expected to show promising results is fine-grained food classification—a task that involves distinguishing closely related food items based on subtle differences in appearance.

Fine-grained food classification is a challenging problem due to high intra-class variability and low inter-class variability among food images [3]. Traditional computer vision algorithms have attempted to address these challenges by incorporating structural enhancements to models or extracting additional features [4–6]. While these methods have achieved some success, they often increase model complexity and may still struggle to capture the nuanced differences inherent in food items.

To overcome these limitations, we propose a novel approach that leverages LLMs to generate descriptive features of food images. By extracting textual descriptions that encapsulate ingredients, cooking methods, and cultural contexts, we enrich the semantic representation of the images. We then fuse these textual features with image features at the feature level using a transformer architecture [7]. This fusion allows the model to consider both visual and contextual information, leading to more accurate and nuanced classifications.

Our method offers several advantages. First, it reduces the need for complex structural modifications to the model by leveraging the rich semantic information from LLMs. Second, the transformer-based fusion effectively combines different data modalities, enhancing the model's ability to generalize across diverse food categories. Experimental results demonstrate that our approach outperforms traditional methods, particularly in distinguishing visually similar food items.

The integration of LLMs into fine-grained food classification not only addresses existing challenges but also opens up new avenues for research in multimodal learning. This



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approach has potential applications in areas such as personalized nutrition, culinary arts, and food delivery services, where accurate food recognition is crucial.

2. Related Work

Fine-grained food classification presents significant challenges in computer vision due to the subtle visual distinctions between food items and the extensive intra-class variations [3]. Traditional classification approaches struggle with subtle differences among food items, especially due to factors like preparation techniques or regional styles.

The integration of large language models (LLMs), such as BERT [8] and GPT-3 [9], into multimodal frameworks has opened new frontiers for fine-grained classification by combining visual and linguistic information. LLMs excel in understanding complex linguistic patterns, providing contextual knowledge that complements visual data. The emergence of vision–language models (VLMs), such as contrastive language–image pretraining (CLIP) [1], has enabled the creation of joint embeddings for visual and textual data, facilitating zero-shot classification and other applications. CLIP has demonstrated strong performance across image recognition tasks by aligning images and text in a shared multimodal space, making it particularly effective for tasks requiring fine-grained differentiation, such as food classification.

Building on advances in multimodal models, MiniGPT-4 [10] bridges the gap between vision and language by generating detailed descriptions of visual content, leveraging a visual encoder and a pretrained language model to enhance fine-grained image classification. This model effectively generates text representations for tasks like image captioning, question-answering, and fine-grained classification. For instance, in food classification, MiniGPT-4 can describe ingredients, preparation methods, and regional distinctions, which helps in distinguishing visually similar dishes through enriched textual context. In this way, multimodal models like MiniGPT-4 not only process visual features but also enhance classification with linguistic information that highlights subtle differences in ingredients or styles.

The use of multimodal prompting in MiniGPT-4, where both text and images are used as input, allows for more accurate semantic representations of fine-grained details. This approach, successfully applied in several classification tasks, improves the model's ability to handle complex visual categories by utilizing knowledge encoded in large-scale language models. Tzelepi and Mezaris [11] demonstrated that combining text embeddings from MiniGPT-4 with visual features from CLIP yields improved classification performance, validating the effectiveness of LLM-based approaches in fine-grained tasks. In the context of fine-grained food classification, where subtle ingredient variations or preparation techniques differentiate food items, these advancements hold substantial implications. LLMs leverage vast textual corpora, capturing domain-specific knowledge about food, including culinary terms and regional cuisines. This enables a nuanced understanding of food-related categories, helping models distinguish between visually similar dishes that differ in subtle but important ways. Moreover, LLMs provide a framework for zero-shot and few-shot learning, enabling the classification of previously unseen food items based on their textual descriptions, further enhancing the model's generalization capabilities.

3. Methods

The overall structure proposed in this paper is shown in Figure 1. The proposed method involves three processes: image feature extraction, text feature extraction, and multimodal learning. The image feature extraction and text feature extraction modules employ preprocessing steps to derive respective image and text embeddings. The multimodal learning module leverages these embeddings to train on combined multimodal features.

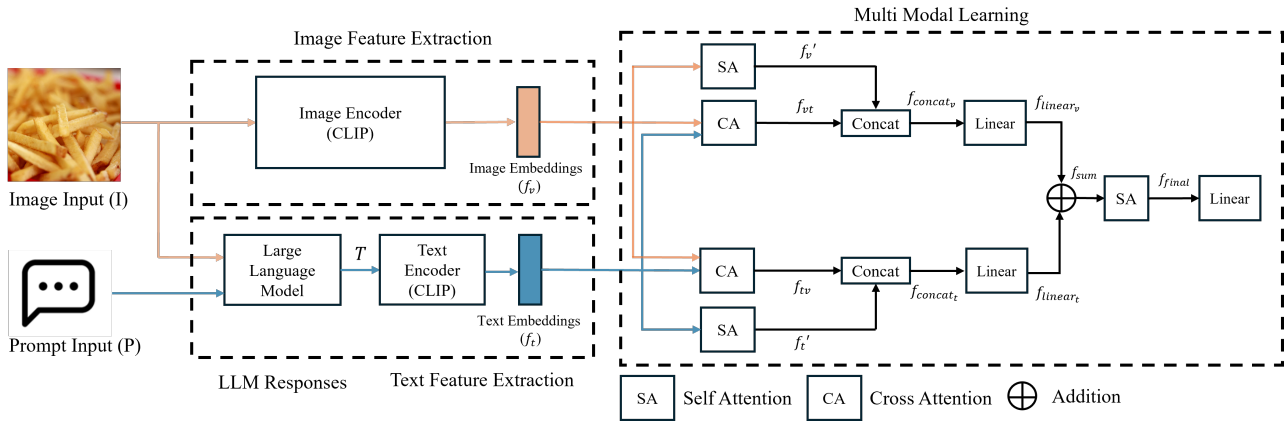


Figure 1. The overall structure of the proposed method.

3.1. Image and Text Feature Extraction

We utilize the contrastive language–image pretraining (CLIP) model, referred to as CLIP [1], for both image and text feature extraction. For image features, the vision component of CLIP, denoted as E_I , processes the image input I through a hybrid architecture combining convolutional and transformer layers to produce high-dimensional visual embeddings f_v :

$$f_v = E_I(I), \quad (1)$$

where I represents the input image data. These embeddings f_v are designed to align semantically with textual data, enabling coherent integration in multimodal applications.

For text features, we employ a pretrained vision–language model (VLM), MiniGPT-4, referred to as LLM [10]. This model generates ten descriptive text captions based on the image input I and the prompt input P :

$$T_1, T_2, \dots, T_{10} = \text{LLM}(I, P) \quad (2)$$

Each text caption T_i is subsequently encoded by the text component of the CLIP model, denoted as E_T , to produce individual text embeddings f_{t_k} :

$$f_{t_k} = E_T(T_k), \quad k = 1, \dots, 10 \quad (3)$$

To combine these embeddings, we perform an average pooling operation over the ten embeddings, $f_{t_1}, f_{t_2}, \dots, f_{t_{10}}$, resulting in a single representative text embedding f_t :

$$f_t = \frac{1}{10} \sum_{k=1}^{10} f_{t_k} \quad (4)$$

These embeddings ensure intrinsic linkage to the visual embeddings f_v , enhancing alignment between the two modalities.

3.2. Multimodal Training

Our multimodal training architecture employs dual pathways to process and integrate the visual embeddings f_v and textual embeddings f_t . This structure enhances the interaction between modalities through distinct self-attention and cross-attention mechanisms.

Each pathway begins with the application of self-attention (SA) mechanisms to the visual embeddings f_v and the textual embeddings f_t . This operation aims to refine the features within each modality by capturing and emphasizing intra-modal relationships, such as spatial dependencies in visual features or contextual relationships in textual data. The resulting refined embeddings are denoted as follows:

$$f'_v = \text{Self-Attention}(f_v) \quad (5)$$

$$f'_t = \text{Self-Attention}(f_t) \quad (6)$$

Following the self-attentions, cross-attention (CA) mechanisms are applied to fuse the features from the two modalities. In one pathway, the cross-attention uses f'_v as the key and value, and f_t as the query

$$f_{tv} = \text{Cross-Attention}(f_t, f'_v, f_v) \quad (7)$$

Conversely, in the second pathway, f_t serves as the key and value, while f'_v acts as the query:

$$f_{vt} = \text{Cross-Attention}(f_v, f'_t, f_t) \quad (8)$$

The outputs from both cross-attention mechanisms are then concatenated with their corresponding self-attention outputs and passed through a linear transformation layer as follows:

$$f_{\text{concat}_v} = \text{Concat}(f'_v, f_{vt}) \quad (9)$$

$$f_{\text{concat}_t} = \text{Concat}(f'_t, f_{tv}) \quad (10)$$

$$f_{\text{linear}_v} = \text{Linear}(f_{\text{concat}_v}) \quad (11)$$

$$f_{\text{linear}_t} = \text{Linear}(f_{\text{concat}_t}) \quad (12)$$

The linearly transformed features are then aggregated into a unified multimodal representation, which is further refined through a final self-attention mechanism to enhance feature interactions.

$$f_{\text{sum}} = f_{\text{linear}_v} + f_{\text{linear}_t} \quad (13)$$

$$f_{\text{final}} = \text{Self-Attention}(f_{\text{sum}}) \quad (14)$$

The final output is then processed through another linear layer to produce the classification. This multimodal training strategy leverages the strengths of each modality's unique features while ensuring that their interactions are maximally informative, leading to more accurate and robust model predictions.

4. Results

4.1. Implementation Details

The performance of the proposed method was evaluated on the food classification problem. We used the public food datasets Food-101 [3] and UEC-Food256 [12]. Food-101 is a real-world food dataset comprising 101 classes collected from a website, with each class containing 750 training images and 250 testing images. UEC-Food256 contains a total of 31,645 images corresponding to 256 food categories from various countries, including France, Vietnam, Japan, and Indonesia. All images were resized so that the longest side is 512 pixels. For both datasets, we used the officially provided training and testing splits. Notably, UEC-Food256 provides five versions of training and testing splits; therefore, performance evaluations were conducted by averaging the results across all five versions.

Performance was assessed using accuracy and F1 score. Accuracy is defined as follows:

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100\%, \quad (15)$$

where N_{correct} is the number of correctly classified samples, and N_{total} is the total number of samples.

The macro F1 score is calculated as follows:

$$\text{Macro F1-score} = \frac{1}{C} \sum_{i=1}^C \text{F1-score}_i, \quad (16)$$

where C is the number of classes, and $F1\ score_i$ is the F1 score for class i .

We used the CLIP model [1] as a feature extractor for images and text, freezing its weights to perform inference only, without any additional training. We utilized the ViT-B/16, ViT-B/32, and ViT-L/14 [13] versions of the CLIP model. For caption generation, we employed MiniGPT-V4 [10] with the Vicuna V0 7B [14] language model. The AdamW optimizer [15] was used to minimize the cross-entropy loss. Other training parameters were fixed: a batch size of 512, 150 epochs, an initial learning rate of 1×10^{-4} , and a cosine annealing learning rate scheduler [16]. All implementations were conducted using the PyTorch framework on a system running Ubuntu 20.04, with 128GB RAM and an NVIDIA V100 GPU with 32GB memory.

4.2. LLM Response in Food Images

To generate descriptive content from food images, we designed a prompt specifically to guide the response of the language model. Given the sensitivity of the output to the phrasing of the prompt, we experimented with several versions and ultimately settled on the following: “Describe what this food image looks like and what ingredients it might contain in sentences. Generate 10 sentences and answer in English”. This prompt was used to input food images into the MiniGPT-4 model, which then generated ten descriptive sentences for each image. Examples of these responses are illustrated in Figure 2a,b.



1. The plate contains a wrap filled with scrambled eggs, cheese, and sautéed vegetables on a bed of crispy potatoes.
2. A side of sliced tomatoes, cucumbers, and carrots are served as a garnish to the dish.
3. On the side is a dipping sauce made of chipotle mayo and a squeeze of lime juice.
4. The blue plate has a bright, bold design that catches the eye.
5. The color scheme gives off a fun, playful vibe that makes you want to dig in.
6. The combination of flavors in this dish is both satisfying and delicious.
7. The presentation is visually appealing and well thought out, making it an impressive dish.
8. The wrap is wrapped tightly around the fillings, holding everything together neatly.
9. The potatoes on the side provide a nice contrast in texture to the softer egg mixture.
10. It's a perfect dish for someone who loves a hearty breakfast or brunch meal.



1. The image shows a display case filled with various types of ice cream in different flavors and colors, including strawberry, chocolate, and vanilla.
2. The ice cream appears to be made from high-quality ingredients, such as fresh fruits and creamy milk.
3. The containers are transparent, allowing customers to see the contents of each flavor.
4. Some of the containers have labels indicating their flavor and price.
5. There are several small scoops of ice cream placed on top of each container for sampling purposes.
6. The display case is lit by fluorescent lights that provide an evenly illuminated view of the products.
7. The background is a clean white, adding contrast to the colorful ice cream.
8. The overall appearance is visually appealing, making customers want to try the various flavors.
9. A sign above the display case reads "Freshly Made Ice Cream" in bold letters, emphasizing the quality of the products.
10. The image captures the essence of a quaint, homemade ice cream shop with its variety of flavors and presentation.

(a)

(b)

Figure 2. Examples of descriptive responses generated by MiniGPT-4 for food images: (a) a response for a ‘breakfast burrito’ image; (b) a response for an ‘ice cream’ image.

4.3. Ablation Study

In this subsection, we present the results of measuring the performance of image-only, text-only, and multimodal inputs on the Food-101 [3] and UEC-Food256 [12] datasets, using three versions of the CLIP model: ViT-B/16, ViT-B/32, and ViT-L/14.

For the Food-101 dataset, the results are summarized in Table 1. We observe that the multimodal approach, which combines both image and text features, consistently outperforms the image-only and text-only methods across all model variants. Specifically, the ViT-L/14 model achieves the highest accuracy of 94.85%, a precision of 94.73%, a recall of 94.90%, and an F1 score of 94.79% when using multimodal inputs. This indicates that integrating textual information generated by LLMs enhances the model’s ability to distinguish between fine-grained food categories.

Comparing the performance of image-only and text-only methods, it is clear that image features contribute more significantly to the classification task than text features alone. The text-only approach yields substantially lower accuracy, precision, recall, and F1 scores, with the highest text-only accuracy being 72.40% for the ViT-B/32 model. This suggests that while textual descriptions provide valuable contextual information, they may lack the discriminative power of visual features in isolation. However, the combination of both modalities leverages the strengths of each, leading to improved overall performance.

Table 1. Ablation study of Food-101. Acc represents accuracy, P represents precision, R represents recall, and F1 represents F1 score.

CLIP Model	Image				Text				Multimodal			
	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
ViT-B/16	91.83	91.63	91.87	91.71	72.38	71.95	72.39	71.94	92.66	92.48	62.62	92.53
ViT-B/32	87.91	87.63	87.88	87.69	72.40	71.99	72.41	71.97	89.92	89.73	89.90	89.77
ViT-L/14	94.39	94.21	94.40	94.28	71.50	71.07	71.55	71.09	94.85	94.73	94.90	94.79

Furthermore, we notice that larger models tend to perform better. The ViT-L/14 model outperforms the smaller ViT-B/16 and ViT-B/32 models across all input modalities and metrics (accuracy, precision, recall, and F1 score). This improvement can be attributed to the increased capacity of the larger model to capture more complex patterns and fine-grained details in the data.

For the UEC-Food256 dataset, the results are presented in Table 2. Similar to the observations on Food-101, the multimodal approach generally achieves higher accuracy and F1 scores compared to image-only and text-only methods. On average, the ViT-L/14 model with multimodal inputs attains the highest accuracy of 80.53% and an F1 score of 83.11%. The improvement over the image-only method is notable, demonstrating the benefit of incorporating textual features in handling a more diverse and complex dataset.

Table 2. Ablation study of UEC-Food256 across all folds.

Fold	CLIP Model	Image		Text		Multimodal	
		Acc (%)	F1 (%)	Acc (%)	F1 (%)	Acc (%)	F1 (%)
Fold 0	ViT-B/16	77.06	80.37	35.78	34.97	78.22	80.85
	ViT-B/32	77.66	80.37	34.91	34.29	74.44	76.83
	ViT-L/14	79.77	82.59	35.16	34.56	79.82	82.50
Fold 1	ViT-B/16	77.04	79.30	33.99	33.42	77.57	79.88
	ViT-B/32	73.62	75.60	35.29	34.62	74.20	76.16
	ViT-L/14	79.93	82.33	33.96	33.35	80.25	82.61
Fold 2	ViT-B/16	77.99	80.28	34.82	33.90	78.08	80.52
	ViT-B/32	73.21	75.19	35.68	35.16	74.17	76.29
	ViT-L/14	80.24	82.82	34.69	34.02	80.67	83.14
Fold 3	ViT-B/16	78.06	80.64	36.13	35.29	78.20	80.70
	ViT-B/32	74.17	76.45	35.77	34.84	75.35	77.82
	ViT-L/14	80.47	83.24	35.33	34.82	81.02	83.71
Fold 4	ViT-B/16	77.29	79.81	35.42	34.22	78.10	80.44
	ViT-B/32	73.80	75.89	35.25	33.95	74.77	76.92
	ViT-L/14	80.01	82.56	34.55	33.56	80.88	83.60
Average	ViT-B/16	77.49	80.08	35.23	34.36	78.03	80.48
	ViT-B/32	74.49	76.70	35.38	34.57	74.59	76.80
	ViT-L/14	80.08	82.71	34.74	34.06	80.53	83.11

The text-only performance on UEC-Food256 is significantly lower than that of image-only and multimodal methods, with average accuracies of around 35%. This is considerably lower than the text-only performance on Food-101, which may be due to the increased complexity and variability in the UEC-Food256 dataset. The dataset includes a wider variety of dishes from multiple countries, making textual descriptions less consistent and more challenging for classification. This indicates that textual descriptions alone are insufficient for accurate classification in this context.

To further investigate the effectiveness of our proposed multimodal method, we evaluated the accuracy of the image model, text model, and our proposed multimodal

model on classes that are visually similar or belong to the same category. Table 3 presents the results of this comparison.

Table 3. Accuracy (%) comparison of image model, text model, and multimodal model on selected visually similar food classes from the UEC-Food 256 dataset (Fold 0).

Group	Class	Image Model	Text Model	Multimodal Model
Burgers	Hamburger	85.37	90.24	95.12
	Sandwiches	89.66	89.66	96.55
Curries	Beef Curry	67.44	48.84	69.77
	Dry Curry	89.47	68.42	94.74
	Yellow Curry	95.00	73.68	100.00
Egg Dishes	Omelet	65.00	0.00	75.00
	Egg Sunny-Side Up	80.00	50.00	85.00
Desserts	Tiramisu	90.00	75.00	95.00
	Rare Cheesecake	85.71	20.00	90.48

The results indicate that our multimodal model consistently outperforms both the image-only and text-only models across these classes. The improvement is especially significant in classes where visual similarity makes discrimination challenging when relying on a single modality. The following are some examples:

- **Similar Fast Foods:** Classes like hamburgers and sandwiches often share visual characteristics. The image model achieves 85.37% and 89.66% accuracy, respectively, while the text model achieves 90.24% and 89.66%. The multimodal model improves classification accuracy further, achieving 95.12% for Hamburger and 96.55% for Sandwiches.
- **Different Types of Curry:** Distinguishing between various curry dishes like beef curry, dry curry, and yellow curry can be difficult based on appearance alone. The image model achieves accuracies ranging from 67.44% to 95.00%, while the text model ranges from 48.84% to 73.68%. The multimodal model enhances accuracy by integrating textual information, achieving up to 100% accuracy for yellow curry.
- **Omelet vs. Egg Sunny-Side Up:** These dishes are visually similar due to their egg-based appearance. The image model achieves 65% and 80% accuracy, respectively, while the text model struggles with 0% and 50% accuracy. The multimodal model, however, improves the accuracy to 75% and 85%, leveraging both visual and textual cues.
- **Desserts:** Dishes like tiramisu and care cheesecake can be challenging to differentiate visually. The image model achieves 90.00% and 85.71% accuracy, while the text model achieves 75.00% and 20.00%. The multimodal model significantly improves accuracy to 95.00% and 90.48%, respectively.

These findings support our claim that the proposed multimodal method is more effective in distinguishing visually similar food items by utilizing complementary information from both modalities.

Overall, the ablation study confirms that the fusion of image and text features via the proposed multimodal approach enhances classification performance on fine-grained food datasets. The integration of LLM-generated textual descriptions with visual features allows the model to capture both visual cues and contextual information, leading to more robust and accurate predictions. These results demonstrate the effectiveness of leveraging multimodal data and underscore the potential of our method in tackling the challenges of fine-grained food classification.

4.4. Comparison with State of the Art

To further evaluate the performance of our proposed approach, we compare it with several state-of-the-art methods on the Food-101 and UEC-Food256 datasets. All methods included in the comparison were trained or fine-tuned on the respective datasets. The results are presented in Tables 4 and 5.

Table 4. Performance comparison with state-of-the-art methods on Food-101.

Method	Accuracy
Swin transformer(tiny) [17]	91.30
Curriculumnet [18]	87.30
Multi-Scale CNN [19]	88.84
PAR-Net [20]	89.3 (1 crop) 90.4 (10 crop)
Grafit [21]	89.50
IELT [22]	90.1
GL-Swin [6]	92.93
Ours(Multimodal, ViT-L/14)	94.85

Table 5. Performance comparison with state-of-the-art methods on UEC-Food256.

Method	Accuracy
WiSeR [23]	83.15
VTnet [24]	83.43
JD-NET [5]	84.00
Swin transformer(tiny) [17]	84.93
GL-Swin [6]	85.78
Ours(Multimodal, ViT-L/14)	80.53

For the Food-101 dataset, our method, utilizing the ViT-L/14 model with multimodal inputs, achieves an accuracy of 94.85%, which is notably higher than all previously reported approaches. For example, GL-Swin [6] records an accuracy of 92.93%. This improvement highlights the effectiveness of integrating LLM-generated textual descriptions with visual features. The enhanced performance suggests that the multimodal fusion effectively leverages contextual information from textual data, enabling the model to better differentiate between fine-grained food categories that may appear visually similar.

Conversely, as shown in Table 5, our approach achieves an accuracy of 80.53% on the UEC-Food256 dataset, which is lower than the accuracies reported by recent methods such as GL-Swin [6] (85.78%) and Swin Transformer (tiny) [17] (84.93%). Several factors could influence this outcome.

- **Dataset Complexity:** The UEC-Food256 dataset includes a broader variety of dishes from multiple countries, introducing greater variability and complexity. This diversity may challenge the consistency and informativeness of the textual descriptions generated by LLMs.
- **Utility of Textual Information:** As observed in our ablation study, the text-only performance on UEC-Food256 is significantly lower compared to Food-101. This indicates that textual descriptions alone provide limited discriminative power for this dataset, potentially reducing the benefits of multimodal fusion.
- **Model Adaptation:** It is possible that the current multimodal fusion strategy is better suited for datasets with less variability. Enhancements to the fusion mechanism or improvements in the quality of textual descriptions could potentially boost performance on more complex datasets like UEC-Food256.

4.5. Discussion

The varying performances across the two datasets highlight both the strengths and limitations of our multimodal approach. While the integration of image and text features results in substantial performance improvements on the Food-101 dataset, it proves less effective for the more diverse UEC-Food256 dataset. This suggests that the benefits of multimodal fusion depend on the characteristics and complexity of the dataset.

Future work could explore the following strategies to address these limitations:

- **Enhanced Text Generation:** improving the quality and consistency of LLM-generated textual descriptions may provide richer contextual information, especially for diverse datasets.
- **Advanced Fusion Techniques:** developing more sophisticated multimodal fusion strategies that can better handle variability and complexity in the data could enhance performance on challenging datasets.
- **Dataset-Specific Adaptations:** tailoring the model architecture or training procedures to accommodate the specific characteristics of different datasets may yield better generalization across various food classification tasks.

5. Conclusions

In this study, we explored the integration of large language models (LLMs) with traditional image classification methods for fine-grained food classification. By leveraging multimodal learning, which combines both visual and textual features, our approach improves the model's ability to differentiate between visually similar food items. This is achieved through the generation of detailed semantic descriptions using MiniGPT-4, which captures essential contextual information related to food ingredients, preparation techniques, and regional variations. These descriptions are then encoded and fused with visual features, enriching the feature set and enhancing the discriminative power of the model.

Our experimental results demonstrate that the proposed method significantly outperforms traditional image-only approaches, particularly in tasks requiring fine-grained differentiation. The transformer-based fusion of image and text features contributes to improved generalization, allowing the model to handle complex food categories more effectively.

The findings of this research offer promising insights into the potential of LLMs in the domain of food classification and open new avenues for future work in multimodal learning. Future research may focus on expanding the dataset, exploring more sophisticated fusion techniques, or applying the methodology to other fine-grained classification tasks beyond food.

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Data Availability Statement: Food-101 and UEC-food256 images' Minigpt-v4 LLM responses are available at https://github.com/kjunhwa/text_response_of_food_image (accessed on 18 November 2024).

Conflicts of Interest: Author Nam-Ho Kim was employed by the company Korean Broadcasting System. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

FGVC	Fine-grained visual classification
LLM	Large language model
VLM	Vision–language model
CLIP	Contrastive language–image pretraining
ViT	Vision transformer
CNN	Convolutional neural network
SOTA	State of the art
GPT	Generative pretrained transformer

BERT	Bidirectional encoder representations from transformers
CA	Cross-attention
SA	Self-attention

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